

On PolarFire SoC (the typical context for HSS, U-Boot, kernel, and filesystem discussions):

1. HSS (Hart Software Services) code

- Stored in: eNVM (Embedded Non-Volatile Memory, 128 KB on-chip).
- HSS is the zero-stage / first-stage bootloader.
- It is programmed into eNVM (via Libero / FlashPro Express job, SoftConsole, or command-line tools).
- On boot (Boot Mode 1 is the most common), the E51 core starts executing from eNVM. HSS typically decompresses itself into L2 Scratchpad and continues running from there.
- Small bare-metal applications can also live entirely in eNVM, but anything larger uses a payload.

2. U-Boot, Linux Kernel, and Filesystem

These are **not** stored in eNVM (it is too small). They are stored in external non-volatile storage and loaded by HSS as an HSS payload.

Component	Typical Storage Location	Notes
U-Boot	<u>eMMC, SD card, or QSPI flash</u> (as part of HSS payload)	Second-stage bootloader. Loaded by HSS into DDR (or sometimes L2).
Linux Kernel (+ Device Tree / FIT image)	<u>Same external media (eMMC / SD / QSPI)</u> as part of the payload or in a separate partition	Loaded by U-Boot into DDR.
Root Filesystem	<u>eMMC or SD card</u> (usually a separate partition, e.g. ext4)	Can also be an initramfs inside the payload for smaller systems.

Typical boot flow (most common Linux setup)

1. Power-on → System Controller → E51 starts HSS from eNVM.
2. HSS initializes hardware (DDR training, clocks, etc.).
3. HSS loads the HSS payload (containing U-Boot, and often the kernel + DTB) from eMMC / SD card / QSPI into DDR.
4. HSS hands off to U-Boot.
5. U-Boot loads the Linux kernel (and DTB) into DDR and boots Linux.

6. Linux mounts the root filesystem from eMMC/SD (or uses initramfs).

Summary of storage

- **eNVM** → HSS (and optionally very small bare-metal apps)
- **sNVM** → Usually not used for HSS/U-Boot/kernel (it is for secure keys, initialization data, or optional secure boot components)
- **eMMC / SD card / QSPI** → U-Boot + Kernel + Filesystem (via HSS payload + partitions)
- **DDR** → Runtime location for U-Boot, kernel, and applications after loading

This is the standard PolarFire SoC software flow used by Microchip reference designs (Icicle Kit etc.)